

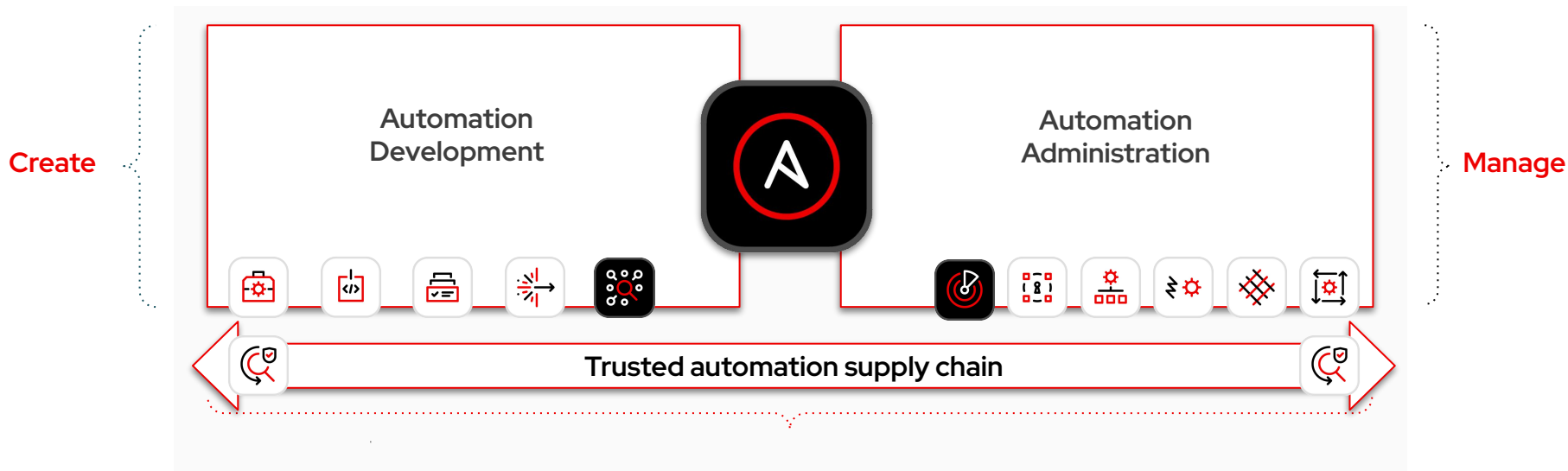
From Create to Deploy: Supercharging the Ansible Automation Content Lifecycle

Tim Appnel
Senior Product Manager
Ansible

Alison Hart
Senior Software Engineer
Ansible

Red Hat Ansible Automation Platform

A unified solution for strategic automation



Create with Ansible Automation Platform

Enable your automation community of practice



Ansible content collections

Set automation projects into motion

Ready-made roles, modules, plug-ins, and templates

Certified and validated for quality assurance and security



Ansible development tools

Suite of supported components

Create, test, and deploy Ansible content



Ansible plug-ins for Red Hat Developer Hub

Curated learning paths, push-button content creation, integrated development tools, and more

Educate new users and boost experienced users



Ansible development workspaces

Opinionated, integrated dev environment

Includes Ansible development tools, VS code integration, and Ansible Lightspeed



Ansible Lightspeed

Collaboration with IBM to bring generative AI to Ansible

Faster and easier Ansible content creation

Ansible development tools

Supported, enterprise-grade components to enable creating, testing, and deployment of Ansible

Create

ansible-creator

Scaffold projects and content

ansible-navigator

Develop and troubleshoot with execution environments

Ansible development environment (ade)

Build and manage virtual environments for content development

Test

Ansible Molecule

Functional test runner for collections, playbooks and roles

ansible-lint

Identify and correct stylistic errors and anti-patterns

pytest-ansible

Added functions for module and plugin Python code unit testing

tox-ansible

Test collections under different Python interpreters and Ansible core versions

Deploy

ansible-builder

Build Ansible execution environments

ansible-sign

Sign and verify Ansible content



Ansible development tools (ansible-dev-tools)

All the tools in one install

What is it?

- ▶ Easy way to install and discover the tools
- ▶ Supported and included with your AAP subscription
- ▶ Available as RPM, Dev Container for VS Code or a Python package

```
$ sudo dnf install ansible-dev-tools
```

```
$ adt --version
```

ansible-builder	3.1.0
ansible-core	2.18.7
ansible-creator	25.8.0
ansible-dev-environment	25.8.0
ansible-dev-tools	25.8.3
ansible-lint	25.8.2
ansible-navigator	25.8.0
ansible-sign	0.1.2
molecule	25.7.0
pytest-ansible	25.8.0
tox-ansible	25.8.0

Ansible dev environments (ade)

Build and manage virtual environments for content development

What is it?

Command line utility for building and managing a virtual environment for Ansible content development

- ▶ Ensures the current development environment is isolated
- ▶ Install all collection python and test requirements
- ▶ Checks for missing system packages
- ▶ Handles all pathing for the current python interpreter

```
ali@-/servicenow.itsm (main) $ ade install -e .\[test] --venv venv -v
Info: Created virtual environment: /home/ali/servicenow.itsm/venv
Info: Installing ansible-core.
Info: Installing local collection from: /home/ali/servicenow.itsm/venv/.ansible-dev-environment/servicenow.itsm/build
Info: File list generated with 'git ls-files'
Note: Installed collections include: servicenow.itsm
Info: Swapping servicenow.itsm with /home/ali/servicenow.itsm
Info: Installing python requirements.
Note: All python requirements are installed.
Info: Checking system packages.
Note: All required system packages are installed.
Note: A virtual environment was specified but has not been activated.
Note: Please activate the virtual environment:
      source venv/bin/activate
ali@-/servicenow.itsm (main) $ source venv/bin/activate
(venv) ali@-/servicenow.itsm (main) $
```

Ansible content creator (**ansible-creator**)

Scaffold projects and content

What is it?

Command line utility for scaffolding Ansible automation projects and contents with leading practices

- ▶ Streamlines collection init process
- ▶ Scaffold collections, playbook projects, execution environments, and more.
- ▶ Add to existing projects or start fresh
- ▶ Integrated with the Ansible VS Code extension

```
$ ansible-creator init collection <collection-name>
$ ansible-creator add resource devfile <dest-path>
$ ansible-creator add plugin filter <plugin-name>
```

Ansible lint (`ansible-lint`)

Identify and correct stylistic errors and anti-patterns

What is it?

- ▶ Command-line tool to lint Ansible playbooks, roles and collections
- ▶ Promote best practices and patterns.
- ▶ Develop consistent code across teams and scale using an opinionated strategy.
- ▶ Integrate into larger development workflows and CI tools.
- ▶ Run transforms using ``ansible-lint --fix`` to easily fix some violations.

```
$ ansible-lint
$ ansible-lint --fix
$ ansible-lint --profile production
$ ansible-lint --format <json, sarif>
$ ansible-lint -x <rules-to-skip>
```


Ansible Molecule (molecule)

Functional test runner for collections, playbooks and roles

What is it?

- ▶ Provides **comprehensive testing suite capabilities**: provision testing environments, manage states, debug test scenarios, and clean up resources .
- ▶ Select, exclude, and verify generated scenarios as desired
- ▶ New **collection-aware functionality** for seamless testing with Ansible content collections
- ▶ New shared state option to manage lifecycle (create/destroy) for **all** test scenarios

```
$ molecule

# Initialize testing scenarios
$ molecule init scenario my-scenario-name

# Test a specific scenario within a component
$ molecule converge -s role1

# Run all tests with a summary report
$ molecule test --all --report

# Use the shared state option
$ molecule test --all --shared-state
```

Ansible Execution Environment builder (`ansible-builder`)

Build reusable, portable runtime environments

What are EEs?

Containerized environments for Ansible automation that **package** Ansible content, dependencies, and runtime requirements.

- ▶ Integrate with private automation hub for **centralized image management**
- ▶ Consistent execution across different infrastructure environments and teams
- ▶ Isolate project dependencies and enable accurate environment testing

```
$ ansible-builder

# Build from current directory
$ ansible-builder build --tag my-ee:latest .

# Use specific base image
$ ansible-builder build --tag my-ee --build-arg
EE_BASE_IMAGE=<image-reference> .

# Inspect collection dependencies
$ ansible-builder introspect --sanitize <collection-path>
```

Ansible content navigator (`ansible-navigator`)

Develop and troubleshoot with execution environments

What is it?

Command line utility and text-based user interface (TUI) for running, testing, and developing Ansible automation content

- ▶ Build, review, and run tests with Ansible execution environments.
- ▶ Develop content and documentation
- ▶ Develop playbooks and troubleshoot problems

```
# Run playbooks
$ ansible-navigator run playbook.yml -i inventory.ini

# Display plugin documentation
$ ansible-navigator doc <plugin>

# Build EEs from your project
$ ansible-navigator builder build --tag my_ansible_ee

# Run tests with execution environments
$ ansible-navigator run test_localhost.yml
--execution-environment-image my_ansible_ee --mode stdout
```

Demo:

Exploring individual development tools



Enhancing the development experience with Visual Studio Code

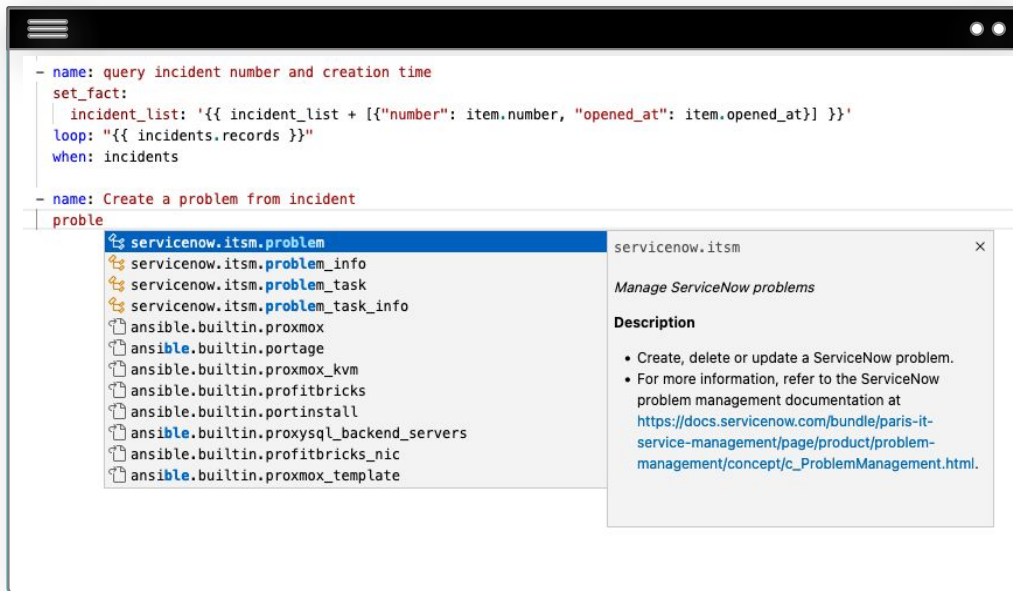
Integrations with Ansible development tools

Ansible VS Code extension

Simplify content creation

What is it?

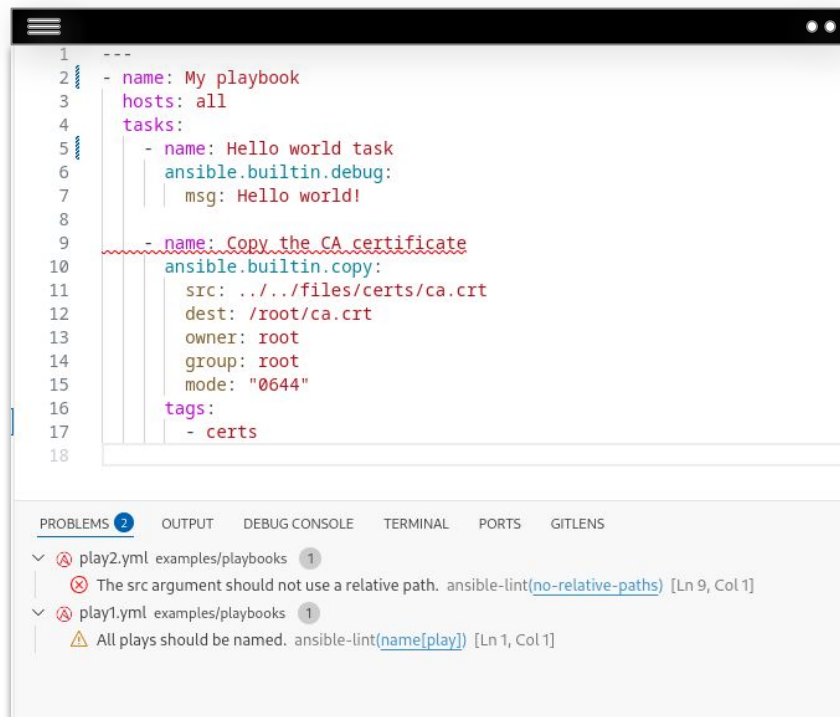
- ▶ Syntax highlighting of keywords
- ▶ Ansible Lightspeed generative AI suggestions
- ▶ Hover and goto functionality
- ▶ Auto-completion on play, block, or task contents, hosts, etc
- ▶ Supports Ansible content development on Windows and Mac OS workstation



Ansible VS Code extension

Simplify content creation

- ▶ Supports writing and running Ansible content with execution environments
- ▶ Integration with ansible-playbook and ansible-navigator for running and troubleshooting playbooks
- ▶ Live validation and feedback from ansible-lint during content creation



The screenshot displays the Ansible VS Code extension interface. The top pane shows a YAML playbook with two tasks. The first task is 'Hello world task' using the 'ansible.builtin.debug' module. The second task is 'Copy the CA certificate' using the 'ansible.builtin.copy' module. The bottom pane shows the 'PROBLEMS' view with two linting errors from ansible-lint:

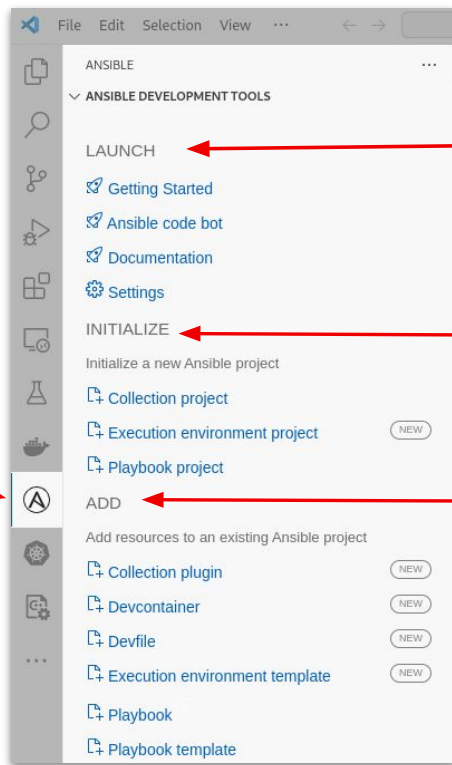
```
1 ---
2 - name: My playbook
3   hosts: all
4   tasks:
5     - name: Hello world task
6       ansible.builtin.debug:
7         msg: Hello world!
8
9     - name: Copy the CA certificate
10       ansible.builtin.copy:
11         src: ../../files/certs/ca.crt
12         dest: /root/ca.crt
13         owner: root
14         group: root
15         mode: "0644"
16       tags:
17         - certs
18
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

- ✖ play2.yml examples/playbooks 1
 - ✖ The src argument should not use a relative path. ansible-lint(no-relative-paths) [Ln 9, Col 1]
- ✖ play1.yml examples/playbooks 1
 - ⚠ All plays should be named. ansible-lint(name[play]) [Ln 1, Col 1]

Ansible VS Code extension

Integrating tools into a unified UI



Access the Ansible extension

Learn more about the extension and Ansible development tools

Initialize projects with ansible-creator

Generate playbooks, add project resources, and build EEs

A screenshot of the 'Create new Ansible collection' form. The form has a title 'Create new Ansible collection' and a subtitle 'Streamlining automation'. It contains fields for 'Namespace' (acme), 'Collection' (acme_tools), and 'Init path' (/root/.ansible/collections/ansible_collections). Below the fields, it shows the 'Collection name' as acme.acme_tools and the 'Project path' as /root/.ansible/collections/ansible_collections/acme/acme_tools. A red arrow points from the text label to the form.

Forms for creating content via the VS Code UI

Normalize and Isolate Development Environments

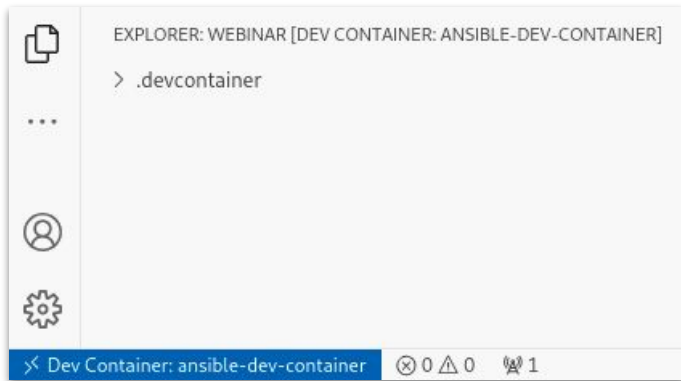
VS Code Ansible Extension + Dev Containers

- ▶ Create containerized development environments with all Ansible development tools included
- ▶ Compatibility with macOS, Windows and Linux
- ▶ Access Microsoft Dev Containers extension through VS Code marketplace



Dev Containers

Microsoft  microsoft.com



Active dev container
in VS Code

Environment Integration with VS Code

VS Code Ansible Extension + Dev Containers

1

INITIALIZE

Initialize a new Ansible project

↳ Collection project

↳ Execution environment project

↳ Playbook project

2

ADD

Add resources to an existing Ansible project

↳ Collection plugin

↳ Devcontainer ←

↳ Devfile

↳ Execution environment template

↳ Playbook

↳ Playbook template

3

Create a devcontainer

Build containerized development environments

Devcontainers are json files used for building containerized development environments.

Enter your project details below to utilize a devcontainer template designed for the Dev Containers extension.

Destination directory *

/workspaces/scaffolded-playbook

Devcontainer path: /workspaces/scaffolded-playbook/.devcontainer

Container image

Downstream (registry.redhat.io/ansible-automation-platform-25/ansible-dev-tools-



Overwrite

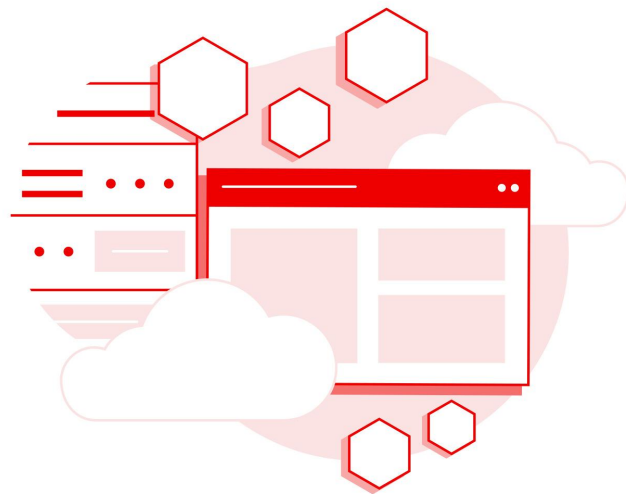
Overwrite an existing devcontainer.

Reset All

Create

Demo:

VS Code and
Dev Container integrations



Connecting Development to Delivery

Automating Build, Test, and Deploy with CI/CD

Automatically enforce consistent style and formatting

`.pre-commit-config.yaml`

repos:

...

- repo: <https://github.com/pre-commit/pre-commit-hooks>

rev: v4.4.0

hooks:

- id: check-merge-conflict

- id: check-symlinks

- id: debug-statements

- id: end-of-file-fixer

- id: no-commit-to-branch

args: [--branch, main]

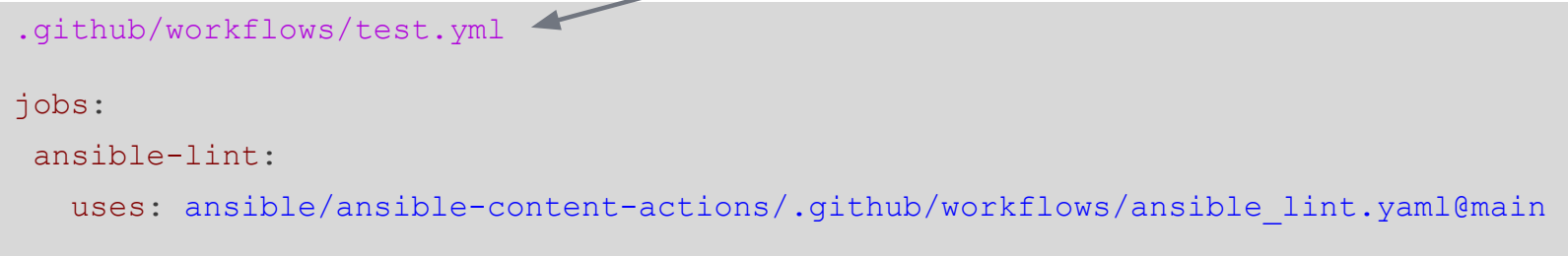
- id: trailing-whitespace

Adds pre-commit hooks for:

- git
- prettier
- isort
- black
- flake8

Continuously validate content

Checks content against Ansible Lint rules.



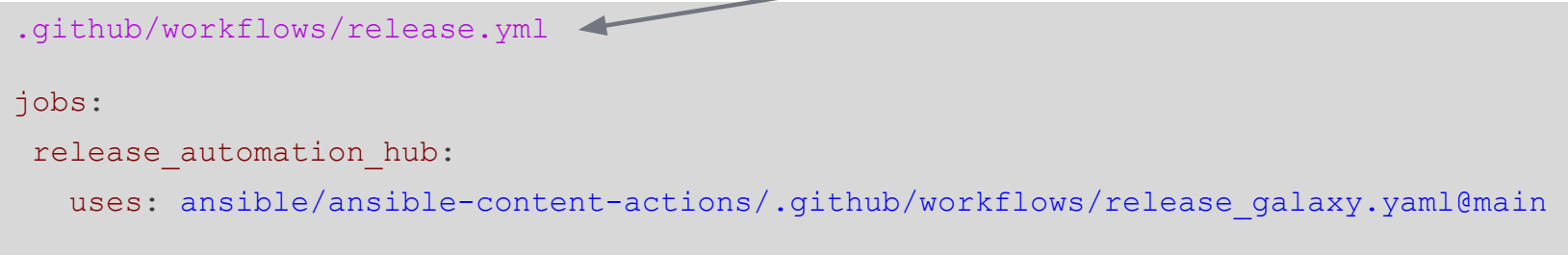
```
.github/workflows/test.yml  
  
jobs:  
  ansible-lint:  
    uses: ansible/ansible-content-actions/.github/workflows/ansible_lint.yml@main
```



Available as a Github action

Fully automate your release pipelines

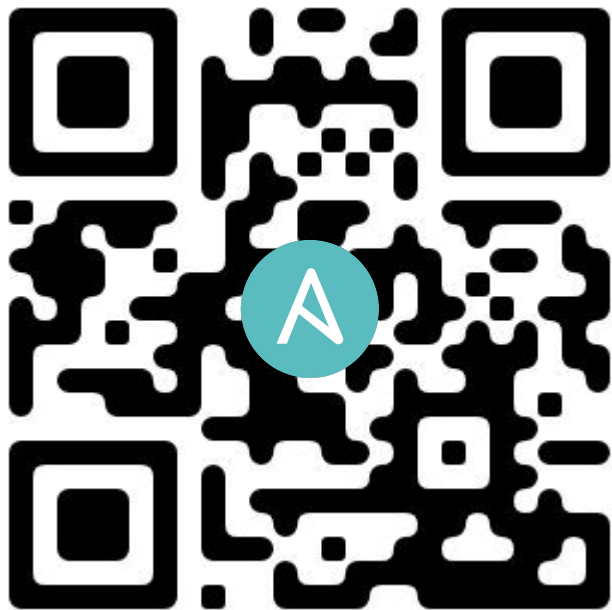
Uploads new releases to Galaxy or Automation Hub.



```
.github/workflows/release.yml  
  
jobs:  
  release_automation_hub:  
    uses: ansible/ansible-content-actions/.github/workflows/release_galaxy.yaml@main
```

Ansible Forum

forum.ansible.com



Ansible DevTools

Hands-on workshop



Thank you

Red Hat is the world's leading provider of enterprise open source software solutions. Award-winning support, training, and consulting services make Red Hat a trusted adviser to the Fortune 500.



linkedin.com/company/red-hat



youtube.com/user/RedHatVideos



facebook.com/redhatinc



twitter.com/RedHat

Standard python testing tool support

- **pytest-ansible**

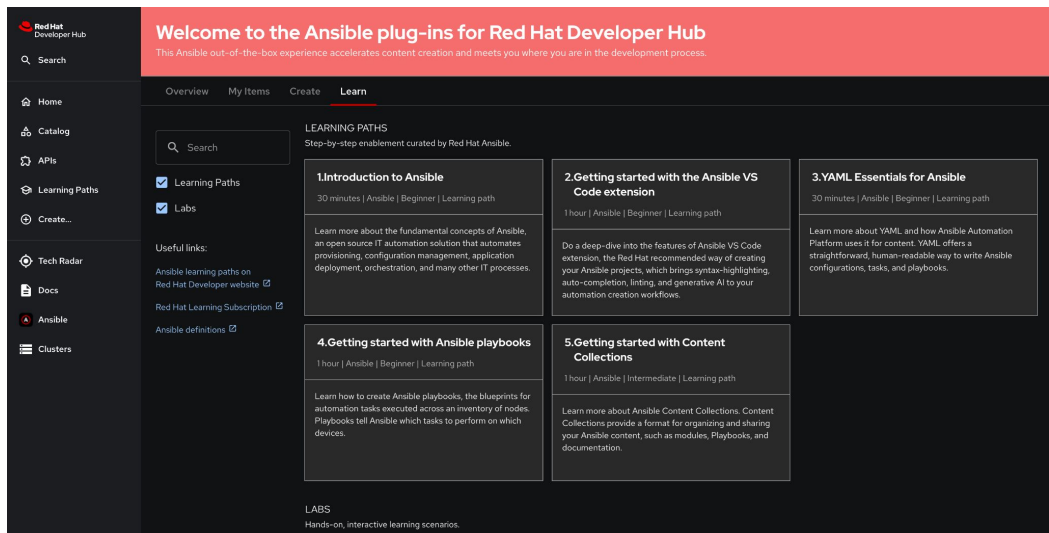
A pytest testing framework extension that provides additional functionality for testing Ansible module and plugin Python code.

- **tox-ansible**

An extension to the tox testing utility that provides additional functionality to check Ansible module and plugin Python code under different Python interpreters and Ansible core versions.

Ansible plug-ins for Red Hat Developer Hub

A unified experience for automation developers



- Ansible-specific portal experience
- Learning paths, labs, and other user onboarding resources
- Push-button Ansible Playbook and collection creation with opinionated best practices
- Integration with OpenShift Dev Spaces (optional)
- RBAC integration with Red Hat Developer Hub

So you can...

- **Onboard** new users faster
- Improve **developer productivity**
- Reduce friction, context switching, and frustration
- Speed **time-to-value** in your automation investment
- Establish governance, control, and **trust**